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Passivity and synchronization of fractional-order coupled neural networks with multiple weights: A PD approach

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ABSTRACT

Traditional control methods for complex neural networks are constrained by their dependence on precise models, neglecting the robustness needed to address high-dimensional coupling characteristics and model uncertainty, which usually results in inflexible responses to disturbances. This paper explores the passivity and synchronization of fractional-order coupled neural networks with multiple weights (FOCNNMWs) using a designed proportional-derivative (PD) control strategy. On one hand, the analysis of various passivity properties in FOCNNMWs, including input-strictly passivity and output-strictly passivity, is conducted through PD controllers, Lyapunov functions, and linear matrix inequality method; on the other hand, a sufficient condition for synchronization is derived by means of output-strictly passivity, utilizing both energy function analysis and the limit properties of Mittag-Leffler function. The proposed PD control approach, via passivity-based analysis, not only avoids the requirement for precise modeling of nonlinear terms but also inherently resists disturbances through passivity's energy dissipation, curbing its impact on synchronization errors. Finally, numerical simulations with two distinct cases confirm the effectiveness of the proposed criteria.

1. Introduction

Recent scholarly investigations have demonstrated the broad applicability of coupled neural networks (CNNs) across multiple domains, including but not limited to civil engineering (Adeli, 2001), natural language processing (Li et al., 2021b), and medical imaging (Yin et al., 2018); moreover, extensive researches have yielded significant advancements in analyzing the dynamic behaviors of CNNs, with synchronization (Chen et al., 2016; Ding & Wang, 2020) emerging as a particularly prominent area of investigation. For example, Ding and Wang (2020) achieved CNN synchronization via an event-triggered intermittent pinning control strategy, leveraging Lyapunov-Krasovskii functional for stability analysis. In Chen et al. (2016), $\mathcal{M}$-matrix-based adaptive synchronization conditions were derived for neutral-type Markovian stochastic CNNs subject to time-varying delays, with mean-square and almost-sure exponential stability properties confirmed by a generalized Lyapunov stability theorem. Additionally, passivity theory, ensuring an input-output energy balance, provides an energy-dissipation framework for analyzing synchronization stability and designing robust controllers, as validated in recent studies (Huang et al., 2018; Lin et al., 2018; Wang et al., 2016). For instance, Lin et al. (2018) investigated the passivity in coupled heterogeneous-dimensional neural networks, where sufficient conditions guaranteeing passivity are developed for both delay-free and time-delayed cases. In a linear matrix inequality (LMI)-based investigation of CNN passivity, Huang et al. (2018) proposed a regrouping method by using communication Laplacian matrix, which overcomes the inherent limitation in error-system approaches that requires full-node information. Wang et al. (2016) not only proposed corresponding passivity definitions for coupled reaction-diffusion neural networks with different input and output dimensions, but also established passivity conditions for both cases with and without time-varying delays; particularly, Dirichlet boundary conditions were employed to simplify the analysis of the reaction-diffusion terms and ensure no energy leakage at the boundaries. However, numerous existing studies including

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the above literature have focused exclusively on structurally simplified models that employ a single shared weight matrix to achieve strong internal coupling within the network. It merits attention that CNNs with multi-weights (CNNMWs) demonstrate significant advantages compared with traditional CNNs, particularly in terms of enhanced fitting accuracy, superior robustness, and higher computational efficiency. Consequently, research on CNNMWs holds greater practical significance in a real controlled system compared to their single-weight counterparts.

Recent studies have extensively explored CNNMWs' dynamic behaviors, particularly the topics of synchronization (Cao et al., 2022; Lin & Liu, 2022; Zhao et al., 2023) and passivity (Wang et al., 2017, 2018). For instance, a chattering-free finite-time synchronization control method was proposed for CNNMWs in Zhao et al. (2023), employing stability theory tools including Lyapunov analysis and integral inequalities. Utilizing a weighted aggretation of normalized left eigenvectors from multiple coupling matrices, Lin and Liu (2022) derived synchronization conditions with adequately large coupling strengths for multi-weighted directly coupled reaction-diffusion neural networks. In Cao et al. (2022), a synchronization control method based on a pinning impulsive protocol was proposed for multi-weighted coupled stochastic reaction-diffusion neural networks with time delays. Regarding passivity, Wang et al. (2017) established multiple Lyapunov-based passivity criteria and analyzed their applications in both delay-free and delayed CNNMWs. Furthermore, Wang et al. (2018) investigated finite-time passivity conditions for CNNMWs with uncertainties under both directed and undirected topologies; the correctness of these conditions was verified using both a three-dimensional undirected CNNMW and a directed CNNMW as examples. However, a notable limitation of most current studies (e.g., Cao et al. (2022), Lin and Liu (2022), Wang et al. (2017, 2018), Zhao et al. (2023)) is their failure in integrating synchronization and passivity into a unified analytical approach. Given the complementary relationship between synchronization and passivity in control theory, a research on both aspects for CNNMWs has significant theoretical and practical value. In this direction, Lin et al. (2020) investigated both passivity and synchronization in delayed CNNMWs composed of non-identical nodes with and without parameter uncertainties, incorporating an event-triggered strategy to ensure control performance while reducing resource consumption; building directly on this research, Wang (2024c) investigated event-triggered passivity and synchronization for multi-derivative coupled reaction-diffusion neural networks, while Huang et al. (2020) addressed the same problem for multi-weighted coupled delayed reaction-diffusion memristive neural networks with fixed and switching topologies, respectively. Progressing further, Tang et al. (2024) not only analyzed finite-time passivity of CNNMWs in both directed and undirected network topologies, but also developed corresponding fixed-time synchronization criteria under output-strict passivity. These studies have significantly advanced current understanding of the dynamical properties in CNNMWs under various network conditions. Similar advances in hybrid control and learning schemes (e.g., for discrete-time cyber-physical systems or singularly perturbed systems, such as Shen et al. (2024b), Shen et al. (2024a)) also provide feasible references for addressing complex control problems in coupled neural networks.

However, it should be noted that the pivotal contributions of Cao et al. (2022), Huang et al. (2020), Lin et al. (2020), Lin and Liu (2022), Tang et al. (2024), Wang et al. (2017, 2018), Wang (2024c), Zhao et al. (2023) for passivity and synchronization analysis remain confined to integer-order CNNMWs. With its inherent capacity to model memory and hereditary properties, fractional calculus enables more precise characterization of complex dynamical behaviors featuring long-range dependence and non-locality. Additional degrees of freedom in modeling and control, afforded by fractional derivatives, enhance strategy flexibility and robustness. Thus, extending passivity and synchronization analyses to fractional-order CNNMWs (FOCNNMWs) emerges as a compelling research direction which both theoretically significant and practically valuable. Related synchronization studies have also been widely explored for fractional-order neural networks, such as quaternion-valued, fuzzy and discrete-time uncertain systems (Li et al., 2021a), Li et al. (2022), Li et al. (2023), providing mature analytical frameworks including fractional differential and difference inequalities for complex system analysis. Currently, there exist a few results regarding the study of passivity and synchronization of FOCNNMWs. For instance, in Liu et al. (2021), the finite-time passivity (including finite-time input-strict and output-strict passivity) of coupled fractional-order neural networks with multi-state or multi-derivative couplings was studied by designing state-feedback and adaptive controllers, with specific fractional-order finite-time passivity definitions. Building on the definitions of finite-time passivity proposed in Liu et al. (2021), Wang (2024a) extended the theoretical framework for fractional-order systems by introducing refined passivity definitions tailored for systems with mismatched input-output dimensions, deriving criteria for multi-weighted fractional-order complex dynamical networks with fixed and adaptive coupling weights, discussing the relationship between output-strict passivity and synchronization, and verifying the reliability of these criteria through numerical examples. In the same vein of exploring passivity in fractional-order systems, in Wang (2024b) examined two types of passivity for multi-adaptive coupled fractional-order reaction-diffusion neural networks and derived output strict passivity criteria using fractional-order differential inequalities, Laplace transform lemmas, and linear matrix inequality tools, while also establishing synchronization conditions for these networks based on the obtained passivity results. Existing approaches to FOCNNMWs, whether relying purely on intrinsic coupling or event-triggered control/feedback control, etc., exhibit inherent limitations: the former lacks active regulation capability, while the latter's discontinuous actuation under disturbances and sensitivity to model inaccuracies degrade robustness compared to continuous control strategies. To tackle these challenges, continuous and robust control strategies, particularly proportional-derivative (PD) control, are crucial for FOCNNMWs, as they offer structural simplicity, real-time responsiveness, and effective disturbance rejection by directly suppressing errors and predicting error trends to enhance synchronization.

In light of the aforementioned analysis, this paper, grounded in the PD control strategy, aims to investigate the passivity and synchronization of FOCNNMWs. This paper's key contributions are presented below.

(1) In contrast to traditional CNNs (e.g., Chen et al. (2016), Ding and Wang (2020), Huang et al. (2018), Lin et al. (2018)) that depend on a single weight matrix, this paper incorporates multi-weight coupling terms into fractional-order CNNs, enabling precise modeling of inter-node coupling effects. Specifically, these terms are summed in dynamic equations, integrated via Kronecker product, and analyzed through LMI.
(2) Unlike some related prior works focusing on integer-order systems (e.g., Liang and Wang (2022), Wang and Zhao (2021), Wang et al. (2022)), a passive and synchronization analysis method for FOCNNMWs is developed in this paper using a PD controller, in which Mittag-Leffler function and its Laplace transform are employed to derive the synchronization criteria.
(3) Some sufficient criteria are derived to guarantee both passivity and synchronization under zero-input conditions.

The layout of this paper is as follows. Section 2 provides fundamental definitions and lemmas pertaining to fractional calculus and passivity analysis. Section 3 conducts the core analyses of passivity and synchronization for the FOCNNMWs. Two numerical simulations are presented in Section 4 to verify the efficacy of the obtained criteria. Finally, Section 5 summarizes the research and outlines potential future work. In the context of this paper, $\mathbb{R}^+$, $\mathbb{R}^n$, $\mathbb{R}^{n\times m}$, $\mathbb{C}$, and $\mathbb{N}$ represent the sets of non-negative real numbers, n-dimentional Euclidean space, $n \times m$ matrices, complex numbers, and natural numbers, respectively. Define $1_\varsigma = (1, 1, \cdots, 1)^T \in \mathbb{R}^\varsigma$. In the FOCNNMWs, $\gamma = 1, 2, \cdots, \varsigma$ stands for the node set, while $\epsilon \subset \gamma \times \gamma$ denotes the undirected connection set. In particular, for a matrix $A \in \mathbb{R}^{n\times n}$, the notation $A \geq 0$ means that A is a semi-positive definite matrix. Finall, the symbol $\otimes$ denotes the Kronecker product.

2. Preliminaries

In preparation for the controller design, key definitions and lemmas are presented first; especially, Caputo fractional calclus and the theory of passivity analysis will be given, which form the basis of the proposed control strategy.

2.1. Basic results of fractional calculus

Given that addressing the fractional differential equations is indispensable in the analysis of FOCNNMWs, this part complies some essential definitions and requisite lemmas of fractional calculus.

Definition 1. Diethelm and Ford (2002) The fractional integral of order $\eth$ is defined as

$$\mathcal{I}_t^{\eth} G(t) = \frac{1}{\Gamma(\eth)} \int_0^t (t-\tau)^{\eth-1} G(\tau) d\tau,$$

with $\Gamma(\eth) = \int_0^{\infty} t^{\eth-1} e^{-\tau} d\tau$.

Throughout this paper, all fractional-order derivatives are considered in the Caputo sense with the order $\eth$ satisfying $0 < \eth < 1$, unless otherwise specified.

Definition 2. Diethelm and Ford (2002) The Caputo fractional derivative of order $\eth$ is expressed as

$$\mathcal{D}_t^{\eth} G(t) = \frac{1}{\Gamma(1-\eth)} \int_0^t \frac{G'(\tau)}{(t-\tau)^{\eth}} d\tau, \tag{1}$$

with $\eth \in (0,1)$. The Laplace transformation of (1) leads to

$$\int_0^{\infty} e^{-st} \mathcal{D}_t^{\eth} G(t) dt = s^{\eth} F(s) - \sum_{k=0}^{n-1} s^{\eth-k-1} G^{(k)}(0).$$

Definition 3. Podlubny (1998) The Mittag-Leffler function is defined by

$$E_{\eta,\delta}(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(k\eta+\delta)},$$

in which $z \in \mathbb{C}$, and $\eta, \delta > 0$. The corresponding expression of Laplace transform is

$$L\{t^{\delta-1} E_{\eta,\delta}(-kt^{\eta})\} = \frac{s^{\eta-\delta}}{s^{\eta}+k},$$

with $k > 0$ being a constant.

Lemma 1. *Podlubny (1998) Suppose that $0 < \eta < 2$, $\delta \in \mathbb{R}$, and $\frac{\eta\pi}{2} < \gamma < \min\{\pi, \eta\pi\}$, then there exists a constant $c > 0$ satisfying*

$$|E_{\eta,\delta}(z)| \leq \frac{c}{1+|z|},$$

for $\gamma \leq |z| \leq \pi$.

Lemma 2. *Podlubny (1998) Suppose that $0 < \eta < 2$, $\frac{\eta\pi}{2} < \gamma < \min\{\pi, \eta\pi\}$, and $\gamma \leq |\arg(z)| \leq \pi$. Then for $n \in \mathbb{N}$ and $\delta \in \mathbb{C}$,*

$$E_{\eta,\delta}(z) = -\sum_{k=1}^{n} \frac{z^{-k}}{\Gamma(\delta-\eta k)} + o\left(\frac{1}{|z|^{1+n}}\right)$$

uniformly holds as $|z| \to \infty$.

Lemma 3. *Li et al. (2009) Consider a smooth function $\boldsymbol{G}(t) : \mathbb{R} \to \mathbb{R}^n$, one has*

$$\mathcal{D}_t^{\eth} \boldsymbol{G}^T(t)\boldsymbol{G}(t) \leq 2\boldsymbol{G}^T(t)\mathcal{D}_t^{\eth}\boldsymbol{G}(t).$$

2.2. Basis for passive analysis

To facilitate the subsequent passivity verification of the investigated network, let us introduce a key definition related to passivity.

Definition 4. Qiu et al. (2023) A fractional-order system with $y \in \mathbb{R}^k$ and input $u \in \mathbb{R}^m$ is said to be:
1. Passive if there exist a positive semi-definite function $V(t)$ and a matrix $_1 \in \mathbb{R}^{k\times m}$ such that

$$\boldsymbol{y}^T S_1 \boldsymbol{u} \geq \mathcal{D}_t^{\eth} V(t).$$

2. Input-strictly passive if there exist a positive semi-definite function $V(t)$, a matrix $S_1 \in \mathbb{R}^{k\times m}$, and a positive definite matrix $S_2 \in \mathbb{R}^{m\times m}$ such that

$$\boldsymbol{y}^T S_1 \boldsymbol{u} - \boldsymbol{u}^T S_2 \boldsymbol{u} \geq \mathcal{D}_t^{\eth} V(t).$$

3. Output-strictly passive if there exist a positive semi-definite function $V(t)$, a matrix $S_1 \in \mathbb{R}^{k\times m}$, and a positive definition matrix $S_3 \in \mathbb{R}^{k\times k}$ such that

$$\boldsymbol{y}^T S_1 \boldsymbol{u} - \boldsymbol{y}^T S_3 \boldsymbol{y} \geq \mathcal{D}_t^{\eth} V(t).$$

Remark 1. The concept of passivity has emerged as a fundamental analytical tool in system stability analysis, providing a robust framework for investigating stabilization problems across various dynamical systems. This approach fundamentally relies on two key mathematical constructs: the storage function $V(t)$, which quantifies the internal energy accumulation within a system, and the supply rate $\boldsymbol{y}^T S_1 \boldsymbol{u}$, representing the external energy injection into the system. Over the past several decades, the passivity-based methodology has attracted substantial research attentions, leading to its successful application in diverse system architectures ranging from classical linear time-invariant systems to complex nonlinear networks. Contemporary research breakthroughs have significantly expanded the theoretical boundaries of passive systems analysis. Notably, recent studies have systematically demonstrated stability preservation in passive network models regardless of dimensional congruence between input and output vectors.

3. Main results

3.1. System description

The present section details the system formulation and establishes key assumptions for controller design and passivity analysis. The considered FOCNNMWs are modeled as

$$\begin{cases} \mathcal{D}_t^{\eth} \boldsymbol{k}_i(t) = -\boldsymbol{D}\boldsymbol{k}_i(t) + \boldsymbol{Q}\boldsymbol{f}(\boldsymbol{k}_i(t)) + \boldsymbol{J} + \boldsymbol{\alpha}_i(t) \\ \qquad\qquad + \boldsymbol{\theta}_i(t) + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m \Phi^m \boldsymbol{k}_j(t), \\ \boldsymbol{\beta}_i(t) = A_1\boldsymbol{\omega}_i(t) + A_2\boldsymbol{\alpha}_i(t), \end{cases} \tag{2}$$

where $i = 1, 2, \cdots, \varsigma$, $m = 1, 2, \cdots, q$; $\boldsymbol{k}_i(t) = [k_{i1}(t), k_{i2}(t), \cdots, k_{in}(t)]^T \in \mathbb{R}^n$ is the state vector of the i−th node; $\boldsymbol{D} = \mathrm{diag}\{d_1, d_2, \cdots, d_n\} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{Q} = (q_{ij})_{n\times n} \in \mathbb{R}^{n\times n}$ are two unknown matrices; $\boldsymbol{f}(\boldsymbol{k}_i(t)) = [f_1(k_{i1}(t)), \cdots, f_n(k_{in}(t))]^T \in \mathbb{R}^n$ is the activation function vector; $\boldsymbol{J} = [\boldsymbol{J}_1, \boldsymbol{J}_2, \cdots, \boldsymbol{J}_n]^T \in \mathbb{R}^n$ is the known constant external input vector; $\boldsymbol{\alpha}_i(t) \in \mathbb{R}^n$ represents the external input; $\boldsymbol{\theta}_i(t) \in \mathbb{R}^n$ and $\boldsymbol{\beta}_i(t) \in \mathbb{R}^p$ stand for the control input and system output, respectively; $A_1 \in \mathbb{R}^{p\times n}$ and $A_2 \in \mathbb{R}^{p\times n}$ are the known matrices; $\boldsymbol{\omega}_i(t) \in \mathbb{R}^n$ is a function vector with respect to $\boldsymbol{k}_i(t)$ that will be given later; $c_m \in \mathbb{R}^+$ indicates the coupling strength of the m−th coupling form; $\Phi^m = \mathrm{diag}\{\phi_1^m, \phi_2^m, \cdots, \phi_n^m\} \in \mathbb{R}^{n\times n}$ is the positive definite inner coupling matrix; the outer coupling matrix $H^m = (h_{ij}^m)_{\varsigma\times\varsigma} \in \mathbb{R}^{\varsigma\times\varsigma}$ satisfies the following property:

$$h_{ij}^m = \begin{cases} h_{ji}^m > 0, & \text{for } (i,j) \in \epsilon, \\ -\sum_{l=1, l\neq i}^{\varsigma} h_{il}^m, & \text{for } i = j, \\ 0, & \text{otherwise}. \end{cases}$$

In order to facilitate the passvity analysis of the network (2), the following assumptions are introduced.

Assumption 1. The activation function $f_i(\cdot) : \mathbb{R} \to \mathbb{R}$, $i = 1, 2, \cdots, n$ satisfies the Lipschitz condition with a known positive constant μ_i:

$$|f_i(x_1) - f_i(x_2)| \le \mu_i |x_1 - x_2|,$$

for any $x_1, x_2 \in \mathbb{R}$.

Assumption 2. All elements of the unknown matrices D and Q obey

$$0 < \underline{d}_i \le d_i \le \overline{d}_i, \quad \underline{q}_{ij} \le q_{ij} \le \overline{q}_{ij}, \quad i = 1, 2, \cdots, n,$$

where $\underline{d}_i, \overline{d}_i, \underline{q}_{ij}$ and $\overline{q}_{ij}$ are knwon constants.

Remark 2. The assumptions introduced above are not restrictive conditions imposed on the proposed PD control strategy, but rather a general and realistic characterization of the inherent uncertainties within the FOCNNMWs themselves. The Lipschitz condition on activation functions is mild and universally satisfied by common neuron models. The interval uncertainty model captures parameter variations within known bounds, which is more practical than requiring precise values. Crucially, the PD controller design does not rely on the exact forms of nonlinearities or parameters; it only requires that the system obeys these broad constraints. Subsequently, the linear matrix inequality conditions derived in Theorems 1–4 serve as verifiable sufficient criteria to rigorously prove that, under such general uncertainties, the simple PD controller can indeed guarantee the desired passivity and synchronization properties. Thus, the assumptions and the LMI framework together establish a robust theoretical foundation that aligns with, rather than contradicts, the practical advantages of the PD control approach.

Lemma 4. Qiu et al. (2023) *Considering that the unknown matrices D and Q that satisfy Assumption 2, there exists a decomposition as*

$$\begin{cases} D = D_0 + M_1\pi_1 N_1, \\ Q = Q_0 + M_2\pi_2 N_2, \end{cases}$$

where $D_0 = \text{diag}\{\frac{\underline{d}_1+\overline{d}_1}{2}, \cdots, \frac{\underline{d}_n+\overline{d}_n}{2}\}$, $Q_0 = \left[\frac{\underline{b}_{ij}+\overline{b}_{ij}}{2}\right]_{n\times n}$, $M_1 = \text{diag}\{\overline{d}_1 - \underline{d}_1, \cdots, \overline{d}_n - \underline{d}_n\}$, $N_1 = M_1$, $N_2 = [\hat{Q}_1, \cdots, \hat{Q}_n]^T$, $\widetilde{q}_i = [\sqrt{\frac{\overline{q}_{i1}-\underline{q}_{i1}}{2}}, \cdots, \sqrt{\frac{\overline{q}_{in}-\underline{q}_{in}}{2}}]^T$, $\hat{Q}_i = \text{diag}\{\sqrt{\frac{\overline{q}_{i1}-\underline{q}_{i1}}{2}}, \cdots, \sqrt{\frac{\overline{q}_{in}-\underline{q}_{in}}{2}}\}$, $M_2 = \text{diag}\{\widetilde{q}_1^T, \cdots, \widetilde{q}_n^T\}$, *and* $\pi_1 \in \mathbb{R}^{n\times n}$, $\pi_2 \in \mathbb{R}^{n^2\times n^2}$ *are unknown function matrices that meet*

$$\pi_1^T\pi_1 \le I_{n\times n}, \quad \pi_2^T\pi_2 \le I_{n^2\times n^2}. \tag{3}$$

Remark 3. The FOCNNMWs (2) discussed in this study adopt the Caputo fractional derivative formulation, which is widely preferred over the Riemann-Liouville definition in control theory applications. This preference arises from the inherent difficulties in extending conventional integer-order systems (as described in Liang and Wang (2022), Wang and Zhao (2021), Wang et al. (2022)) to their fractional-order counterparts when using Riemann-Liouville derivatives, primarily due to the complex limit formulations required for initial condition determination. In contrast, Caputo-type fractional-order systems retain the same straightforward initial value requirements as their integer-order counterparts, significantly simplifying analysis and implementation, especially when the fractional order ð lies within the interval $(0, 1)$.

Remark 4. The selection of PD control in this study arises from a balanced consideration of the dynamic complexity inherent in multi-weighted coupling structures, the demands of theoretical rigor in analysis, and the practical requirements of implementation. Compared to proportional-integral (PI) control, PD avoids potential issues such as delayed response and integral windup under noisy or highly nonlinear conditions, making it particularly suitable for scenarios that emphasize dynamic performance and oscillation suppression. In contrast to pure proportional control, the inclusion of a derivative term introduces necessary phase lead and enhanced damping, which helps coordinate interactions among multiple coupling paths and reduces oscillatory behavior. Furthermore, unlike adaptive or event-triggered schemes that depend on online updates or complex logic, the fixed structure of PD control integrates more seamlessly into analytical frameworks built on Lyapunov functionals and matrix inequalities, thereby facilitating the derivation of explicit and verifiable algebraic criteria. Thus, PD control not only provides structural simplicity and ease of tuning, but also aligns effectively with passivity theory, offering a tractable and theoretically grounded approach for analyzing complex networks with multi-weight couplings.

3.2. Passivity analysis of FOCNNMWs

Set $\widetilde{\boldsymbol{k}}(t) = \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{k}_\rho(t)$, which yields

$$\begin{aligned}
\mathcal{D}_t^{\eth}\widetilde{\boldsymbol{k}}(t) &= \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \mathcal{D}_t^{\eth}\boldsymbol{k}_\rho(t) \\
&= \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} [-D\boldsymbol{k}_\rho(t) + Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) + J + \boldsymbol{\alpha}_\rho(t) \\
&\quad + \boldsymbol{\theta}_\rho(t) + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{\rho j}^m \Phi^m \boldsymbol{k}_j(t)] \\
&= -\frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} D\boldsymbol{k}_\rho(t) + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} J \\
&\quad + \frac{1}{\varsigma}\sum_{\rho}^{\varsigma} \boldsymbol{\alpha}_\rho(t) + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\theta}_\rho(t) \\
&\quad + \frac{1}{\varsigma}\sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m\left(\sum_{\rho}^{\varsigma} H_{\rho j}^m\right)\Phi^m \boldsymbol{k}_j(t) \\
&= -D\widetilde{\boldsymbol{k}}(t) + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) + J + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\alpha}_\rho(t) \\
&\quad + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\theta}_\rho(t).
\end{aligned}$$

Define $\boldsymbol{\omega}_i(t) = \boldsymbol{k}_i(t) - \widetilde{\boldsymbol{k}}(t)$, one obtains

$$\begin{aligned}
\mathcal{D}_t^{\eth}\boldsymbol{\omega}_i(t) &= \mathcal{D}_t^{\eth}\boldsymbol{k}_i(t) - \mathcal{D}_t^{\eth}\widetilde{\boldsymbol{k}}(t) \\
&= -D\boldsymbol{k}_i(t) + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) + J + \boldsymbol{\alpha}_i(t) + \boldsymbol{\theta}_i(t) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m \Phi^m \boldsymbol{k}_j(t) + D\widetilde{\boldsymbol{k}} - J \\
&\quad - \frac{1}{\varsigma}\sum_{\rho}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) - \frac{1}{\varsigma}\sum_{\rho}^{\varsigma} \boldsymbol{\alpha}_\rho(t) - \frac{1}{\varsigma}\sum_{\rho}^{\varsigma} \boldsymbol{\theta}_\rho(t) \\
&= -D\boldsymbol{\omega}_i(t) + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m \Phi^m (\widetilde{\boldsymbol{k}}(t) + \boldsymbol{\omega}_j(t)) \\
&\quad + \boldsymbol{\alpha}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\alpha}_\rho(t) + \boldsymbol{\theta}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\theta}_\rho(t) \\
&= -D\boldsymbol{\omega}_i(t) + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) \\
&\quad + \sum_{m=1}^{q} c_m(\sum_{j=1}^{\varsigma} H_{ij}^m)\Phi^m \widetilde{\boldsymbol{k}}(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\alpha}_\rho(t) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m \Phi^m \boldsymbol{\omega}_j(t) + \boldsymbol{\alpha}_i(t) + \boldsymbol{\theta}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} \boldsymbol{\theta}_\rho(t)
\end{aligned}$$

$$
\begin{aligned}
&= -D\boldsymbol{\omega}_i(t) + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m \boldsymbol{\omega}_j + \boldsymbol{\alpha}_i(t) \\
&\quad - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\alpha}_\rho(t) + \boldsymbol{\theta}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\theta}_\rho(t).
\end{aligned} \tag{4}
$$

To guarantee the passivity and synchronization in the FOCNNMWs (2), the PD controller is devise as

$$
\boldsymbol{\theta}_i(t) = \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} H_{ij}^m \Phi^m\left(b_D^m \mathcal{D}_t^{\eth}\boldsymbol{k}_j(t) + b_p^m \boldsymbol{k}_j(t)\right), \tag{5}
$$

where $i = 1, 2, \cdots, \varsigma$, $b_D^m \in \mathbb{R}^+$, and $b_p^m \in \mathbb{R}^+$.

By performing an equivalent transformation on (5), one obtains

$$
\begin{aligned}
\boldsymbol{\theta}_i(t) &= \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} H_{ij}^m \Phi^m\left(b_D^m \mathcal{D}_t^{\eth}\boldsymbol{k}_j(t) + b_p^m \boldsymbol{k}_j(t)\right) \\
&= \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} H_{ij}^m \Phi^m\Big[b_D^m\left(\mathcal{D}_t^{\eth}\boldsymbol{\omega}_j(t) + \mathcal{D}_t^{\eth}\widetilde{\boldsymbol{k}}(t)\right) \\
&\qquad + b_p^m\left(\boldsymbol{\omega}_j(t) + \widetilde{\boldsymbol{k}}(t)\right)\Big] \\
&= \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} H_{ij}^m \Phi^m\left(b_D^m \mathcal{D}_t^{\eth}\boldsymbol{\omega}_j(t) + b_p^m \boldsymbol{\omega}_j(t)\right).
\end{aligned} \tag{6}
$$

Let $\hat{\boldsymbol{\mu}} = \mathrm{diag}\{\mu_1^2, \mu_2^2, \cdots, \mu_n^2\}$, $\boldsymbol{\alpha}(t) = [\boldsymbol{\alpha}_1^T(t), \boldsymbol{\alpha}_2^T(t), \cdots, \boldsymbol{\alpha}_\varsigma^T(t)]^T$, $\boldsymbol{\omega}(t) = [\boldsymbol{\omega}_1^T(t), \boldsymbol{\omega}_2^T(t), \cdots, \boldsymbol{\omega}_\varsigma^T(t)]^T$, $\boldsymbol{\beta}(t) = [\boldsymbol{\beta}_1^T(t), \boldsymbol{\beta}_2^T(t), \cdots, \boldsymbol{\beta}_\varsigma^T(t)]^T$, and $\boldsymbol{F}(\boldsymbol{k}(t)) = [\boldsymbol{f}^T(\boldsymbol{k}_1(t)), \boldsymbol{f}^T(\boldsymbol{k}_2(t)), \cdots, \boldsymbol{f}^T(\boldsymbol{k}_\varsigma(t))]^T$. The passivity of the network (2) will be proven via the subsequent theorem.

Theorem 1. *For the FOCNNMWs (2) complying with Assumptions 1–2, if there exists a matrix $Z \in \mathbb{R}^{p\varsigma\times n\varsigma}$ fulfilling*

$$
\begin{bmatrix} \Xi_1 & \Psi_1^T \\ \Psi_1 & \Xi_2 \end{bmatrix} \le 0, \tag{7}
$$

in which $\Xi_1 = I_\varsigma \otimes (-2D_0 + M_1M_1^T + N_1^TN_1 + Q_0Q_0^T + M_2M_2^T + \hat{\boldsymbol{\mu}} + \hat{\boldsymbol{\mu}}N_2^TN_2) + 2\sum_{m=1}^{q}\left(c_m + b_p^m\right)(H^m \otimes \Phi^m)$, $\Psi_1 = I_{\varsigma n} - Z^T\left(I_\varsigma \otimes A_1\right)$, and $\Xi_2 = -\left(I_\varsigma \otimes A_2^T\right)Z - Z^T\left(I_\varsigma \otimes A_2\right)$, then the network (2) is passive under the PD controller (5).

Proof. By substituting (6) into (4), one has

$$
\begin{aligned}
\mathcal{D}_t^{\eth}\boldsymbol{\omega}_i(t) &= -D\boldsymbol{\omega}_i(t) + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m\Phi^m \boldsymbol{\omega}_j + \boldsymbol{\alpha}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\alpha}_\rho(t) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} H_{ij}^m \Phi^m\left(b_D^m \mathcal{D}_t^{\eth}\boldsymbol{\omega}_j(t) + b_p^m \boldsymbol{\omega}_j(t)\right) \\
&\quad - \frac{1}{\varsigma}\sum_{\rho}^{\varsigma}\boldsymbol{\theta}_\rho(t).
\end{aligned}
$$

Define a Lyapunov function candidate for the FOCNNMWs (2) as

$$
V(t) = \sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\boldsymbol{\omega}_i(t) - \sum_{m=1}^{q} b_D^m \boldsymbol{\omega}^T(t)(H^m \otimes \Phi^m)\boldsymbol{\omega}(t), \tag{8}
$$

where $\boldsymbol{\omega}(t) = [\boldsymbol{\omega}_1^T(t), \boldsymbol{\omega}_2^T(t), \cdots, \boldsymbol{\omega}_\varsigma^T(t)]^T$.

According to Lemmas 3 and 4, one obtains

$$
\begin{aligned}
\mathcal{D}_t^{\eth}V(t) &\le 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\mathcal{D}_t^{\eth}\boldsymbol{\omega}_i(t) \\
&\quad - 2\sum_{m=1}^{q} b_D^m\boldsymbol{\omega}^T(t)(H^m \otimes \Phi^m)\mathcal{D}_t^{\eth}\boldsymbol{\omega}(t) \\
&= 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)[-D\boldsymbol{\omega}_i(t) + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} c_m H_{ij}^m\Phi^m\boldsymbol{\omega}_j(t) + \boldsymbol{\alpha}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\alpha}_\rho(t) \\
&\quad + \sum_{m=1}^{q}\sum_{j=1}^{\varsigma} H_{ij}^m\Phi^m\left(b_D^m\mathcal{D}_t^{\eth}\boldsymbol{\omega}_j(t) + b_p^m\boldsymbol{\omega}_j(t)\right) \\
&\quad - \frac{1}{\varsigma}\sum_{\rho}^{\varsigma} Q\boldsymbol{f}(\boldsymbol{k}_\rho(t)) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\theta}_\rho(t)] \\
&\quad - 2\sum_{m=1}^{q} b_D^m\boldsymbol{\omega}^T(t)(H^m \otimes \Phi^m)\mathcal{D}_t^{\eth}\boldsymbol{\omega}(t) \\
&= -2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)D\boldsymbol{\omega}_i(t) + 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)Q\boldsymbol{f}(\boldsymbol{k}_i(t)) \\
&\quad - 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)Q\boldsymbol{f}(\widetilde{\boldsymbol{k}}(t)) + 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\boldsymbol{\alpha}_i(t) \\
&\quad + 2\sum_{m=1}^{q}\sum_{i=1}^{\varsigma}\sum_{j=1}^{\varsigma} c_m H_{ij}^m\boldsymbol{\omega}_i^T(t)\Phi^m\boldsymbol{\omega}_j(t) \\
&\quad + \sum_{m=1}^{q}\sum_{i=1}^{\varsigma}\sum_{j=1}^{\varsigma} b_p^m H_{ij}^m\boldsymbol{\omega}_i^T(t)\Phi^m\boldsymbol{\omega}_j(t) \\
&\quad + 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)Q\left(\boldsymbol{f}(\widetilde{\boldsymbol{k}}(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{f}(\boldsymbol{k}_\rho(t))\right) \\
&\quad - \sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\left(\frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\alpha}_\rho(t) + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\theta}_\rho(t)\right).
\end{aligned} \tag{9}
$$

In addition, based on $\boldsymbol{\omega}_i(t) = \boldsymbol{k}_i(t) - \widetilde{\boldsymbol{k}}(t)$, it holds that

$$
\begin{aligned}
\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i(t) &= \sum_{i=1}^{\varsigma}\left(\boldsymbol{k}_i(t) - \widetilde{\boldsymbol{k}}(t)\right) \\
&= \sum_{i=1}^{\varsigma}\left(\boldsymbol{k}_i(t) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{k}_\rho(t)\right) \\
&= \sum_{i=1}^{\varsigma}\boldsymbol{k}_i(t) - \sum_{i=1}^{\varsigma}\left(\frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{k}_\rho(t)\right) \\
&= \sum_{i=1}^{\varsigma}\boldsymbol{k}_i(t) - \sum_{\rho=1}^{\varsigma}\boldsymbol{k}_\rho(t) = 0.
\end{aligned}
$$

Therefore, the following two equations hold, i.e.,

$$
\begin{aligned}
&\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)Q\left(\boldsymbol{f}(\widetilde{\boldsymbol{k}}(t)) - \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{f}(\boldsymbol{k}_\rho(t))\right) = 0, \\
&\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\left(\frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\alpha}_\rho(t) + \frac{1}{\varsigma}\sum_{\rho=1}^{\varsigma}\boldsymbol{\theta}_\rho(t)\right) = 0.
\end{aligned} \tag{10}
$$

Furthermore, one can get

$$
2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)Q\left(\boldsymbol{f}_i(\boldsymbol{k}_i(t)) - \boldsymbol{f}(\widetilde{\boldsymbol{k}}(t))\right) \le \sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\left(Q^TQ + \hat{\boldsymbol{\mu}}\right)\boldsymbol{\omega}_i(t). \tag{11}
$$

It follows from (9), (10) and (11) that

$$
\begin{aligned}
\mathcal{D}_t^{\eth}V(t) &= -2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)D\boldsymbol{\omega}_i(t) + 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}_i^T(t)\boldsymbol{\alpha}_i(t) \\
&\quad + 2\sum_{i=1}^{\varsigma}\boldsymbol{\omega}^T(t)Q\left(\boldsymbol{f}(\boldsymbol{k}_i(t) - \boldsymbol{f}(\widetilde{\boldsymbol{k}}(t))\right)
\end{aligned}
$$

$$+\sum_{m=1}^{q}\sum_{i=1}^{\varsigma}\sum_{j=1}^{\varsigma}\left(c_m+b_p^m\right)H_{ij}^m\omega_i^T(t)\Phi^m\omega_j(t)$$
$$\leq -2\sum_{i=1}^{\varsigma}\omega_i^T(t)\big(D_0+M_1\pi_1N_1\big)\omega_i(t)$$
$$+2\sum_{i=1}^{\varsigma}\omega_i^T(t)\big(Q_0+M_2\pi_2N_2\big)\Big(f(\boldsymbol{k}_i(t)-f(\widetilde{\boldsymbol{k}}(t))\Big)$$
$$+2\sum_{m=1}^{q}\sum_{i=1}^{\varsigma}\sum_{j=1}^{\varsigma}\left(c_m+b_p^m\right)H_{ij}^m\omega_i^T(t)\Phi^m\omega_i(t)$$
$$+2\sum_{i=1}^{\varsigma}\omega_i^T(t)\alpha_i(t).$$

According to Young's inequality, Assumption 1, and (3), one has

$$\mathcal{D}_t^{\vartheta}V(t)\leq +\sum_{i=1}^{\varsigma}\omega_i^T(t)\big(M_1M_1^T+N_1^TN_1\big)\omega_i(t)$$
$$-2\sum_{i=1}^{\varsigma}\omega_i^T(t)D_0\omega_i(t)+2\sum_{i=1}^{\varsigma}\omega_i^T(t)\alpha_i(t)$$
$$+\sum_{i=1}^{\varsigma}\omega_i(t)\big(Q_0Q_0^T+\hat{\mu}I_n\big)\omega_i(t)$$
$$+\sum_{i=1}^{\varsigma}\omega_i^T(t)\big(M_2M_2^T+\hat{\mu}N_2^TN_2\big)\omega_i(t)$$
$$+2\sum_{m=1}^{q}\sum_{i=1}^{\varsigma}\sum_{j=1}^{\varsigma}\left(c_m+b_p^m\right)H_{ij}^m\omega_i^T(t)\Phi^m\omega_j(t)$$
$$=\sum_{i=1}^{\varsigma}\omega_i^T(t)(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T$$
$$+\hat{\mu}I_n+M_2M_2^T+\hat{\mu}N_2^TN_2)\omega_i(t)$$
$$+2\sum_{m=1}^{q}\sum_{i=1}^{\varsigma}\sum_{j=1}^{\varsigma}\left(c_m+b_p^m\right)H_{ij}^m\omega_i^T(t)\Phi^m\omega_j(t)$$
$$+2\sum_{i=1}^{\varsigma}\omega_i^T(t)\alpha_i(t)$$
$$=\omega^T(t)[I_\varsigma\otimes(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T$$
$$+M_2M_2^T+\hat{\mu}+\hat{\mu}N_2^TN_2)]\omega(t)+2\omega^T(t)\alpha(t)$$
$$+2\sum_{m=1}^{q}\left(c_m+b_p^m\right)\omega^T(t)(H^m\otimes\Phi^m)\omega(t). \tag{12}$$

Then, according to (12) one has

$$\mathcal{D}_t^{\vartheta}V(t)-2\beta^T(t)Z\alpha(t)$$
$$\leq\omega^T(t)[I_\varsigma\otimes(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T+M_2M_2^T$$
$$+\hat{\mu}+\hat{\mu}N_2^TN_2)]\omega(t)-2\alpha^T(t)\big(I_\varsigma\otimes A_2^T\big)Z\alpha(t)$$
$$+2\sum_{m=1}^{q}\left(c_m+b_p^m\right)\omega^T(t)(H^m\otimes\Phi^m)\omega(t)+2\omega^T(t)\alpha(t)$$
$$-2\omega^T(t)\big(I_\varsigma\otimes A_1^T\big)Z\alpha(t)$$
$$=\omega^T(t)\big[I_\varsigma\otimes(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T+M_2M_2^T$$
$$+\hat{\mu}+\hat{\mu}N_2^TN_2)+2\sum_{m=1}^{q}\left(c_m+b_p^m\right)(H^m\otimes\Phi^m)]\omega(t)$$
$$+2\omega^T(t)\alpha(t)-\omega^T(t)[(I_\varsigma\otimes A_1^T)Z+Z^T(I_\varsigma\otimes A_1)]\alpha(t)$$
$$-\alpha^T(t)[(I_\varsigma\otimes A_2^T)Z+Z^T(I_\varsigma\otimes A_2)]\alpha(t) \tag{13}$$
$$=\omega^T(t)[I_\varsigma\otimes(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T+M_2M_2^T$$
$$+\hat{\mu}+\hat{\mu}N_2^TN_2)+2\sum_{m=1}^{q}(c_m+b_p^m)(H^m\otimes\Phi^m)]\omega(t)$$
$$+\omega^T(t)\big[2I_{\varsigma n}-(I_\varsigma\otimes A_1^T)Z-Z^T(I_\varsigma\otimes A_1)\big]\alpha(t)$$
$$+\alpha^T(t)\big[-(I_\varsigma\otimes A_2^T)Z-Z^T(I_\varsigma\otimes A_2)\big]\alpha(t)$$
$$=\boldsymbol{k}^T(t)\begin{bmatrix}\Xi_1 & \Psi_1^T\\ \Psi_1 & \Xi_2\end{bmatrix}\boldsymbol{k}(t)\leq 0,$$

in which $\boldsymbol{k}(t)=[\omega^T(t),\alpha^T(t)]^T$. It follows from (13) and Definition 4 that the network (2) is passive. □

Building upon the proof methodology of Theorem 1, a sufficient condition guaranteeing that the network (2) is input-strictly passive is established in the following Theorem.

Theorem 2. *If there exist two matrices $Z\in\mathbb{R}^{p\varsigma\times n\varsigma}$ and $0<E_1\in\mathbb{R}^{n\varsigma\times n\varsigma}$ satisfying*

$$\begin{bmatrix}\Xi_1 & \Psi_1^T\\ \Psi_1 & \Xi_3\end{bmatrix}\leq 0, \tag{14}$$

where $\Xi_3=\Xi_2+2E_1$, then the network (2) achieves input-strict passivity under the PD control (5).

Proof. Application of (13) yields

$$\mathcal{D}_t^{\vartheta}V(t)-2\beta^T(t)Z\alpha(t)+2\alpha^T(t)E_1\alpha(t)$$
$$=\omega^T(t)[I_\varsigma\otimes(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T$$
$$+M_2M_2^T+\hat{\mu}+\hat{\mu}N_2^TN_2)$$
$$+2\sum_{m=1}^{q}(c_m+b_p^m)(H^m\otimes\Phi^m)]\omega(t)$$
$$+\omega^T(t)[I_{\varsigma n}-Z^T(I_\varsigma\otimes A_1)]\alpha(t)$$
$$+\alpha^T(t)[I_{\varsigma n}-(I_\varsigma\otimes A_1^T)Z]\omega(t)+\alpha^T(t)[-(I_\varsigma\otimes A_2^T)Z$$
$$-Z^T(I_\varsigma\otimes A_2)+2E_1\big]\alpha(t)$$
$$=\boldsymbol{k}^T(t)\begin{bmatrix}\Xi_1 & \Psi_1^T\\ \Psi_1 & \Xi_3\end{bmatrix}\boldsymbol{k}(t)$$
$$\leq 0,$$

which means that the network (2) achieves input-strict passivity under the PD controller (5). □

Analogous to Theorem 1, the following holds:

Theorem 3. *Given two matrices $Z\in\mathbb{R}^{p\varsigma\times n\varsigma}$ and $0<E_2\in\mathbb{R}^{p\varsigma\times p\varsigma}$ such that*

$$\begin{bmatrix}\Xi_4 & \Psi_2^T\\ \Psi_2 & \Xi_5\end{bmatrix}\leq 0, \tag{15}$$

where $\Xi_4=I_\varsigma\otimes(-2D_0+M_1M_1^T+N_1^TN_1+Q_0Q_0^T+M_2M_2^T+\hat{\mu}+\hat{\mu}N_2^TN_2)+\sum_{m=1}^{q}(c_m+b_p^m)(H^m\otimes\Phi^m)+2(I_\varsigma\otimes A_1^T)E_2(I_\varsigma\otimes A_1)$, $\Psi_2=I_\varsigma-(I_\varsigma\otimes A_1^T)Z+2(I_\varsigma\otimes A_2^T)E_2(I_\varsigma\otimes A_1)$, $\Xi_5=-(I_\varsigma\otimes A_2^T)Z-Z^T(I_\varsigma\otimes A_2)+2(I_\varsigma\otimes A_2^T)E_2(I_\varsigma\otimes A_2)$, then the network (2) achieves output-strict passivity under the PD control (5).

Proof. Implementation of (13) leads to

$$
\begin{aligned}
&\mathcal{D}_t^{\eth} V(t) - 2\boldsymbol{\beta}^T(t) Z \boldsymbol{\alpha}(t) + 2\boldsymbol{\beta}^T(t) E_2 \boldsymbol{\beta}(t) \\
&\leq \boldsymbol{\omega}^T(t)[I_\varsigma \otimes (-2D_0 + M_1 M_1^T + N_1^T N_1 + Q_0 Q_0^T \\
&\quad + M_2 M_2^T + \hat{\boldsymbol{\mu}} + \hat{\boldsymbol{\mu}} N_2^T N_2) + \sum_{m=1}^{q} (c_m + b_{p^m})(H^m \otimes \Phi^m) \\
&\quad + 2(I_\varsigma \otimes A_1^T) E_2 (I_\varsigma \otimes A_1)]\boldsymbol{\omega}(t) + \boldsymbol{\omega}^T(t)[I_{\varsigma n} \\
&\quad - Z^T(I_\varsigma \otimes A_1) + 2(I_\varsigma \otimes A_1^T) E_2 (I_\varsigma \otimes A_2)]\boldsymbol{\alpha}(t) \\
&\quad + \boldsymbol{\alpha}^T(t)[I_{\varsigma n} - (I_\varsigma \otimes A_1^T) Z + 2(I_\varsigma \otimes A_2^T) E_2 (I_\varsigma \otimes A_1)]\boldsymbol{\omega}(t) \\
&\quad + \boldsymbol{\alpha}^T(t)[-Z^T(I_\varsigma \otimes A_2) + 2(I_\varsigma \otimes A_2^T) E_2 (I_\varsigma \otimes A_2) \\
&\quad - (I_\varsigma \otimes A_2^T) Z]\boldsymbol{\alpha}(t) \\
&= \boldsymbol{k}^T(t) \begin{bmatrix} \Xi_4 & \Psi_2^T \\ \Psi_2 & \Xi_5 \end{bmatrix} \boldsymbol{k}(t) \\
&\leq 0,
\end{aligned}
$$

which means that the network (2) achieves output-strict passivity via the PD control (5). □

3.3. Synchronization analysis of FOCNNMWs

For the FOCNNMWs (2), the definition of synchronization and sufficient condition for synchronization will be given below.

Definition 5. When $\boldsymbol{\alpha}_i = 0$ for $i = 1, 2, \cdots, \varsigma$, the network (2) achieves synchronization if

$$
\lim_{t \to +\infty} ||\boldsymbol{k}_i(t) - \widetilde{\boldsymbol{k}}(t)|| = \lim_{t \to +\infty} ||\boldsymbol{\omega}_i(t)|| = 0, \qquad i = 1, 2, \cdots, \varsigma.
$$

Theorem 4. *If $A_1^T A_1 > 0$ and there exist two matrices $Z \in \mathbb{R}^{p\varsigma \times p\varsigma}$ and $0 < E_2 \in \mathbb{R}^{p\varsigma \times p\varsigma}$ such that*

$$
\begin{bmatrix} \Xi_4 & \Psi_2^T \\ \Psi_2 & \Xi_5 \end{bmatrix} \leq 0, \tag{16}
$$

then the FOCNNMWs (2) is synchronized under zero input via the PD controller (5).

Proof. Theorem 3 implies

$$
\mathcal{D}_t^{\eth} V(t) \leq 2\boldsymbol{\beta}^T(t) Z \boldsymbol{\alpha}(t) - 2\boldsymbol{\beta}^T(t) E_2 \boldsymbol{\beta}(t).
$$

Under $\boldsymbol{\alpha}(t) = \mathbf{0}$, one obtains

$$
\mathcal{D}_t^{\eth} V(t) \leq -2\boldsymbol{\beta}^T(t) E_2 \boldsymbol{\beta}(t),
$$

and

$$
\boldsymbol{\beta}(t) = (I_\varsigma \otimes A_1)\boldsymbol{\omega}(t) + (\varsigma \otimes A_2)\boldsymbol{\alpha}(t) = (I_\varsigma \otimes A_1)\boldsymbol{\omega}(t).
$$

Then, the following result holds, i.e.,

$$
\begin{aligned}
\mathcal{D}_t^{\eth} V(t) &\leq -2[(I_\varsigma \otimes A_1)\boldsymbol{\omega}(t)]^T E_2 [(I_\varsigma \otimes A_1)\boldsymbol{\omega}(t)] \\
&= -2\boldsymbol{\omega}^T(t)(I_\varsigma \otimes A_1^T) E_2 (I_\varsigma \otimes A_1)\boldsymbol{\omega}(t) \\
&\leq -2\lambda_1 ||\boldsymbol{\omega}(t)||^2,
\end{aligned} \tag{17}
$$

where λ_1 is the minimum eigenvalue of the positive definite matrix $(I_\varsigma \otimes A_1^T) E_2 (I_\varsigma \otimes A_1)$. By performing inequality scaling on $V(t)$ [see (8)], one has

$$
\begin{aligned}
V(t) &= ||\boldsymbol{\omega}(t)||^2 - \sum_{m=1}^{q} b_D^m \boldsymbol{\omega}^T(t)(H^m \otimes \Phi^m)\boldsymbol{\omega}(t) \\
&\leq ||\boldsymbol{\omega}(t)||^2 - \sum_{m=1}^{q} b_D^m \lambda_2 ||\boldsymbol{\omega}(t)||^2 \\
&= (1 - \sum_{m=1}^{q} b_D^m \lambda_2) ||\boldsymbol{\omega}(t)||^2,
\end{aligned} \tag{18}
$$

where λ_2 is the minimum eigenvalue of the semi-negative definite matrix $H^m \otimes \Phi^m$. It follows from (17) and (18) that

$$
\mathcal{D}_t^{\eth} V(t) \leq -2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} V(t).
$$

Performing inequality scaling on $V(t)$ [see (8)] again yields

$$
\begin{aligned}
V(t) &= ||\boldsymbol{\omega}(t)||^2 - \sum_{m=1, m \neq \kappa}^{q} b_D^m \boldsymbol{\omega}^T(t)(H^m \otimes \Phi^m)\boldsymbol{\omega}(t) \\
&\quad - b_D^\kappa \boldsymbol{\omega}^T(t)(H^\kappa \otimes \Phi^\kappa)\boldsymbol{\omega}(t) \\
&\geq ||\boldsymbol{\omega}(t)||^2 - b_D^\kappa \boldsymbol{\omega}^T(t)(H^\kappa \otimes \Phi^\kappa)\boldsymbol{\omega}(t) \\
&\geq ||\boldsymbol{\omega}(t)||^2 - b_D^\kappa l_2 \boldsymbol{\omega}^T(t)(I_\varsigma \otimes \Phi^\kappa)\boldsymbol{\omega}(t) \\
&\geq ||\boldsymbol{\omega}(t)||^2 - b_D^\kappa l_2 \lambda_3 ||\boldsymbol{\omega}(t)||^2 \\
&= (1 - b_D^\kappa l_2 \lambda_3) ||\boldsymbol{\omega}(t)||^2,
\end{aligned} \tag{19}
$$

where l_2 denotes the maximum non-zero eigenvalue of semi-negative definite matrix H^κ, and λ_3 is the minimum eigenvalue of the positive definite matrix $I_\varsigma \otimes \Phi^\kappa$.

Therefore, considering Eqs. (18) and (19), it can be inferred that

$$
(1 - b_D^\kappa l_2 \lambda_3) ||\boldsymbol{\omega}(t)||^2 \leq V(t) \leq (1 - \sum_{m=1}^{q} b_D^m \lambda_2) ||\boldsymbol{\omega}(t)||^2.
$$

Obviously, there is a function $\rho(t) > 0$ fulfilling

$$
\mathcal{D}_t^{\eth} V(t) = -2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} V(t) - \rho(t).
$$

Taking the Laplace transform of both sides yields

$$
s^{\eth} V(s) - s^{\eth} V(0) = -2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} V_1(s) - \Lambda(s), \tag{20}
$$

where $V(s)$ and $\Lambda(s)$ denote the Laplace transforms of $V(t)$ and $\rho(t)$, respectively. By equivalent transformation of (20), one can obtain

$$
V(s) = \frac{s^{\eth - 1}}{s^{\eth} + 2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2}} V(0) - \frac{1}{s^{\eth} + 2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2}} \Lambda(s). \tag{21}
$$

Performing the inverse transform on (21) yields

$$
\begin{aligned}
V(t) &= E_{\eth,1}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} t^{\eth}) V(0) \\
&\quad - t^{\eth - 1} E_{\eth,\eth}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m} t^{\eth}) * \rho(t),
\end{aligned}
$$

where $*$ represents convolution operator. It follows from the non-negativity of $\rho(t)$ and the Mittag-Leffler function $E_{\eth,\eth}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m} t^{\eth})$ that $t^{\eth - 1} E_{\eth,\eth}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m} t^{\eth}) * \rho(t) \geq 0$. Thus,

$$
\begin{aligned}
&(1 - b_D^\kappa l_2 \lambda_3) ||\boldsymbol{\omega}(t)||^2 \leq V(t) \leq \\
&E_{\eth,1}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} t^{\eth}) V(0).
\end{aligned}
$$

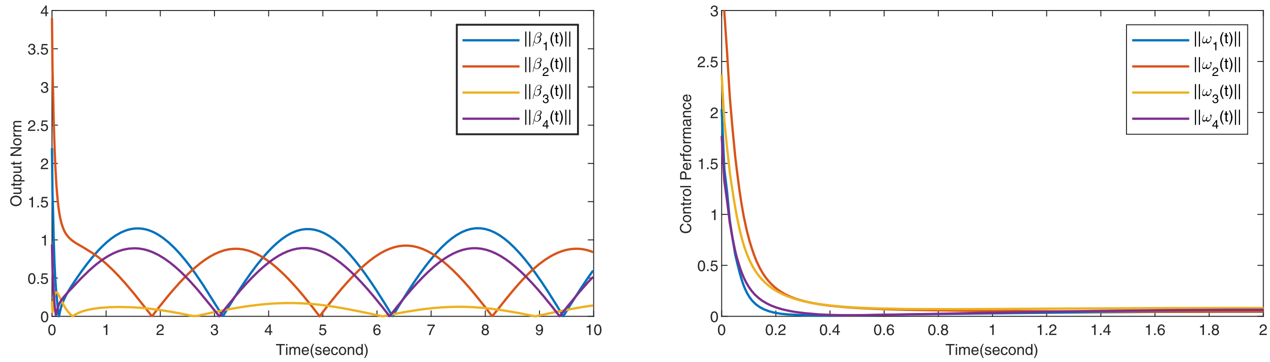
Through equivalent transformation, one has

$$
||\boldsymbol{\omega}(t)||^2 \leq \frac{V(0)}{1 - b_D^\kappa l_2 \lambda_3} E_{\eth,1}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} t^{\eth}).
$$

It follows from $\lim_{t \to +\infty} E_{\eth,1}(-2 \frac{\lambda_1}{1 - \sum_{m=1}^{q} b_D^m \lambda_2} t^{\eth}) = 0$ that $\lim_{t \to +\infty} ||\boldsymbol{\omega}(t)||^2 = 0$.

Therefore, the network (2) achieves synchronization. □

Remark 5. Building upon the earlier work Qiu et al. (2023) which focused on analyzing the intrinsic passivity of autonomous networks, this study designs a PD control strategy that actively ensures and exploits



(a) (b)

Fig. 1. Simulation results. (a) Neuron output norms of the network (22) under the external inputs $\boldsymbol{\alpha}_i(t)$, $i = 1, 2, 3, 4$, via the PD controller (5). (b) Evolutions of $||\boldsymbol{\omega}_i(t)||$, $i = 1, 2, 3, 4$, in the network (22) under zero input via the PD controller (5).

these properties to achieve synchronization in a more general multi-weighted network model. The work shifts the focus from passive analysis to control synthesis, providing explicit PD control laws that enforce and accelerate synchronization under zero input–a result with enhanced practical applicability. Furthermore, the framework is extended from standard couplings to multi-weighted topologies, capturing more complex and realistic interconnections. The PD controller is formulated to handle this increased architectural complexity, thereby broadening the applicability of the theory. Importantly, while the analysis is developed for PD control, the constructed framework is intentionally structured to highlight its transferability to other advanced strategies, such as PI or event-triggered control, positioning the present PD-based results as a foundational case for future research. Thus, this work bridges the gap between passivity theory and controlled synchronization in fractional-order networks, offering both theoretical novelty and a practical design methodology.

4. Simulation

The established theorems are numerically verified via two representative scenarios.

4.1. Example 1

The following FOCNNMWs are considered:

$$\begin{cases} D_t^{\eth}\boldsymbol{k}_i(t) = -Dk_i + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) + J + \boldsymbol{\alpha}_i(t) + \boldsymbol{\theta}_i(t) \\ \qquad + 7\sum_{j=1}^{4} H_{ij}^1\Phi^1\boldsymbol{k}_j(t) + 7\sum_{j=1}^{4} H_{ij}^2\Phi^2\boldsymbol{k}_j(t) \\ \qquad + 7\sum_{j=1}^{4} H_{ij}^3\Phi^3\boldsymbol{k}_j(t), \\ \boldsymbol{\beta}_i(t) = A_1\boldsymbol{\omega}_i(t) + A_2\boldsymbol{\alpha}_i(t), \qquad i = 1, 2, 3, 4, \end{cases} \tag{22}$$

where $\eth = 0.98$, $\underline{D} = \mathrm{diag}\{1.8, 1.9, 2.0\}$, $\overline{D} = \mathrm{diag}\{1.82, 1.92, 2.02\}$, $M_1 = N_1 = \mathrm{diag}\{0.02, 0.02, 0.02\}$, $\pi_1 = \sin t \cdot I_{3\times 3}$, $\pi_2 = \sin t \cdot I_{9\times 9}$, $J = [0.2, 0.1, 0.6]^T$, $f_r(\chi) = 0.25 \cdot (|\chi + 1| - |\chi - 1|)$, $r = 1, 2, 3$, $\boldsymbol{\alpha}_1(t) = [\sin t, \sin t, \sin t]^T$, $\boldsymbol{\alpha}_2(t) = [0.2\sin t, \cos t, \sin t]^T$, $\boldsymbol{\alpha}_3(t) = [\cos t, 0, \sin t]^T$, $\boldsymbol{\alpha}_4(t) = [\sin t, \sin t, 0.3\cos t]^T$, $\boldsymbol{k}_1(t) = [0, 0, 0]^T$, $\boldsymbol{k}_2(t) = [1, 2, 5]^T$, $\boldsymbol{k}_3(t) = [2, 3, 0]^T$, $\boldsymbol{k}_4(t) = [1, 0, 0]^T$, $\boldsymbol{\theta}_i(t) = [\theta_{i1}(t), \theta_{i2}(t), \theta_{i3}(t)]^T \in \mathbb{R}^3$, $\beta_i(t) \in \mathbb{R}$, $A_1 = [1.2, 0, 0.8]$, $A_2 = [0, 0.9, 1.2]$,

$$\underline{Q} = \begin{bmatrix} 0.2 & 0.3 & 0.4 \\ 0.24 & 0.67 & 0.78 \\ 0.35 & 0.57 & 0.98 \end{bmatrix}, \quad \overline{Q} = \begin{bmatrix} 0.22 & 0.32 & 0.42 \\ 0.26 & 0.69 & 0.8 \\ 0.37 & 0.59 & 1 \end{bmatrix},$$

$$M_2 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.1 & 0.1 & 0.1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0.1 & 0.1 \end{bmatrix},$$

$$N_2 = \begin{bmatrix} 0.1 & 0 & 0 & 0.1 & 0 & 0 & 0.1 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0.1 & 0 & 0 & 0.1 & 0 \\ 0 & 0 & 0.1 & 0 & 0 & 0.1 & 0 & 0 & 0.1 \end{bmatrix}^T,$$

$$H^1 = \begin{bmatrix} -0.5 & 0.1 & 0.2 & 0 \\ 0.1 & -0.8 & 0 & 0.5 \\ 0.2 & 0 & -0.6 & 0 \\ 0 & 0.5 & 0 & -1 \end{bmatrix}, \Phi^1 = \begin{bmatrix} 1.2 & 0 & 0 \\ 0 & 1.1 & 0 \\ 0 & 0 & 1.3 \end{bmatrix},$$

$$H^2 = \begin{bmatrix} -0.6 & 0.3 & 0.1 & 0 \\ 0.3 & -0.5 & 0 & 0.1 \\ 0.1 & 0 & -0.6 & 0 \\ 0 & 0.1 & 0 & -0.3 \end{bmatrix}, \Phi^2 = \begin{bmatrix} 0.8 & 0 & 0 \\ 0 & 1.2 & 0 \\ 0 & 0 & 0.9 \end{bmatrix},$$

$$H^3 = \begin{bmatrix} -0.9 & 0.3 & 0.4 & 0 \\ 0.3 & -0.5 & 0 & 0.1 \\ 0.4 & 0 & -0.9 & 0 \\ 0 & 0.1 & 0 & -0.3 \end{bmatrix}, \Phi^3 = \begin{bmatrix} 1.1 & 0 & 0 \\ 0 & 0.9 & 0 \\ 0 & 0 & 1.2 \end{bmatrix}.$$

Taking $\mu_r = 0.5$, it is obvious that

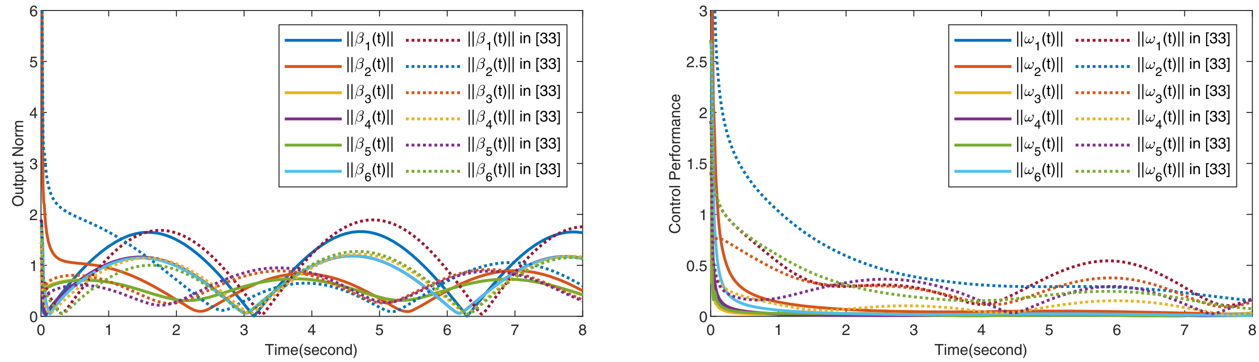
$$|f_r(\chi_1) - f_r(\chi_2)| \le 0.5 \cdot |\chi_1 - \chi_2|, \tag{23}$$

for any $\chi_1, \chi_2 \in \mathbb{R}$.

The parameters of the PD controller are set $b_P^1 = b_P^2 = b_P^3 = 0.1$. At the same time, by using the MATLAB YALMIP Toolbox, one can obtain a feasible solution for the LMI (7), i.e.,

$$Z = \begin{bmatrix} -0.0093 & 0.0970 & 0.0478 & 0.0475 \\ 0.2430 & 0.2022 & 0.0362 & 0.1539 \\ 0.0493 & 0.1596 & 0.0715 & 0.0468 \\ 0.1106 & 0.0665 & -0.0194 & 0.0591 \\ 0.2007 & 0.3980 & 0.0545 & 0.1376 \\ 0.1557 & 0.1236 & 0 & 0.1251 \\ 0.0371 & -0.0293 & 0.0223 & 0.0857 \\ 0.0330 & 0.0576 & 0.1803 & 0.1402 \\ 0.0891 & 0 & 0.0363 & 0.0662 \\ 0.0310 & 0.0746 & 0.0639 & 0.0334 \\ 0.1616 & 0.1386 & 0.1361 & 0.2771 \\ 0.0382 & 0.1136 & 0.0918 & 0.1132 \end{bmatrix}^T.$$

Then, the passivity of the network (22) is established under the PD cotroller (5). Similarly, taking $b_P^1 = b_P^2 = b_P^3 = 0.1$ and using the MATLAB YALMIP Toolbox, one can obtain a set of feasible solutions for the



(a) (b)

Fig. 2. Comparative simulations with and without PD control. (a) Change curves of $||\boldsymbol{\beta}_i(t)||$, $i = 1,2,3,4,5,6$. (b) Evolutions of $||\boldsymbol{\omega}_i(t)||$, $i = 1,2,3,4,5,6$.

LMI (14), i.e.

$$Z = \begin{bmatrix} 0.0169 & 0.1508 & 0.0746 & 0.0570 \\ 0.2348 & 0.2062 & 0.2463 & -0.0107 \\ 0.2727 & 0.0221 & 0.2309 & -0.1239 \\ 0.1666 & -0.0205 & 0.0432 & 0.2017 \\ 0.1937 & 0.2212 & 0.0238 & 0.2695 \\ 0.0402 & 0.1976 & -0.0358 & 0.1712 \\ 0.1339 & 0.0749 & 0 & -0.0498 \\ 0.2228 & 0.0134 & 0.4137 & -0.1408 \\ 0.2511 & -0.0228 & 0.0975 & -0.0705 \\ 0.0719 & 0.1819 & -0.0230 & 0.0296 \\ -0.0119 & 0.2836 & -0.1549 & 0.3612 \\ -0.1290 & 0.1481 & -0.0462 & 0.2818 \end{bmatrix}^T,$$

and

$$E_1 = I_4 \otimes \begin{bmatrix} 0.3037 & 0.0507 & 0.1990 \\ 0.0507 & 0.0329 & 0.0812 \\ 0.1990 & 0.0812 & 0.2436 \end{bmatrix}.$$

Then, the input-strict passivity of the network (22) is achieved via the PD control (5).

Finally, if the controller parameters are set to $b_P^1 = b_P^2 = b_P^3 = 0.1$, and $b_D^1 = b_D^2 = b_D^3 = 0.1$, then a set of feasible solutions for the LMI (15) are solved as

$$Z = \begin{bmatrix} 0.5496 & 0.0152 & 0.0203 & 0 \\ 0.1086 & 0.0374 & 0.0303 & 0.0091 \\ 0.4162 & 0.0397 & 0.0402 & 0.0078 \\ 0.0159 & 0.5485 & 0 & 0.0193 \\ 0.0383 & 0.1179 & 0.0073 & 0.0373 \\ 0.0411 & 0.4245 & 0.0062 & 0.0476 \\ 0.0183 & 0 & 0.5559 & 0 \\ 0.0291 & 0.0079 & 0.0977 & 0.0023 \\ 0.0369 & 0.0065 & 0.4140 & 0.0015 \\ 0 & 0.0193 & 0 & 0.5536 \\ 0.0116 & 0.0376 & 0.0022 & 0.1178 \\ 0.0087 & 0.0475 & 0.0014 & 0.4274 \end{bmatrix}^T,$$

and

$$E_2 = \begin{bmatrix} 0.0196 & 0.0051 & 0.0054 & 0.0024 \\ 0.0051 & 0.0219 & 0.0022 & 0.0086 \\ 0.0054 & 0.0022 & 0.0172 & 0 \\ 0.0024 & 0.0086 & 0 & 0.0214 \end{bmatrix},$$

which means that the output-strict passivity of the network (22) is achieved via the PD controller (5). As observed in Fig. 1(a), even under the external inputs $\boldsymbol{\alpha}_i(t)$, $i = 1,2,3,4$, the system's output norm fluctuation amplitude remains bounded ($||\boldsymbol{\beta}_i(t)|| < 1.5$, $i = 1,2,3,4$) and exhibits a decaying trend due to the disturbance suppression effect of the PD controller (5); these dual characteristics, namely the decaying property and boundedness of output energy, further confirm the system's output-strictly passivity.

Select $b_D^1 = b_D^2 = b_D^3 = 0.1$. Then, it is straightforward to verify that $A_1^T A_1 > 0$ and (16) hold. Furthermore, Fig. 1(b) demonstrates that $\lim_{t\to+\infty} ||\boldsymbol{\omega}_i(t)|| = 0$, $i = 1,2,3,4$, which, combined with Theorem 4, ultimately confirms that the FOCNNMWs (22) achieve synchronization without input via the designed PD controller (5).

4.2. Example 2

In this simulation case, the efficacy of the proposed PD control strategy is evaluated through comparative analysis with Liu and Wang (2021)'s baseline results (obtained without implementing any control framework). For fair benchmarking, identical system models are maintained, with the proposed PD controller being the sole differentiating factor.

Consider the following FOCNNMWs,

$$\begin{cases} D_t^{\eth}\boldsymbol{k}_i(t) = -D k_i + Q\boldsymbol{f}(\boldsymbol{k}_i(t)) + J + O\boldsymbol{\alpha}_i(t) + \boldsymbol{\theta}_i(t) \\ \qquad + 0.1\sum_{j=1}^{6} H_{ij}^1 \Phi^1 \boldsymbol{k}_j(t) + 0.3\sum_{j=1}^{6} H_{ij}^2 \Phi^2 \boldsymbol{k}_j(t) \\ \qquad + 0.2\sum_{j=1}^{6} H_{ij}^3 \Phi^3 \boldsymbol{k}_j(t), \\ \boldsymbol{\beta}_i(t) = A_1\boldsymbol{\omega}_i(t) + A_2\boldsymbol{\alpha}_i(t), \qquad i = 1,2,3,4,5,6, \end{cases} \tag{24}$$

where $\eth = 0.8$, $D = \mathrm{diag}\{0.7, 0.9, 0.8\}$, $J = [0.7, 0.6, 0.8]^T$, $f_r(\chi) = 0.25 \cdot (|\chi + 1| - |\chi - 1|)$, $r = 1,2,3$, $\boldsymbol{\alpha}_1(t) = [\sin t, \sin t, \sin t]^T$, $\boldsymbol{\alpha}_2(t) = [0.2\sin t, \cos t, \sin t]^T$, $\boldsymbol{\alpha}_3(t) = [\cos t, 0, \sin t]^T$, $\boldsymbol{\alpha}_4(t) = [\sin t, \sin t, 0.3\cos t]^T$, $\boldsymbol{\alpha}_5(t) = [\cos t, 0, \sin t]^T$, $\boldsymbol{\alpha}_6(t) = [\sin t, \sin t, 0.3\cos t]^T$, $\boldsymbol{k}_1(t) = [0,0,0]^T$, $\boldsymbol{k}_2(t) = [10,2,8]^T$, $\boldsymbol{k}_3(t) = [2,0,0]^T$, $\boldsymbol{k}_4(t) = [1,0,0]^T$, $\boldsymbol{k}_5(t) = [1,0,0]^T$, $\boldsymbol{k}_6(t) = [0,0,0]^T$, $\boldsymbol{\theta}_i(t) = [\theta_{i1}(t), \theta_{i2}(t), \theta_{i3}(t)]^T \in \mathbb{R}^3$, $\boldsymbol{\beta}_i(t) \in \mathbb{R}^{4\times 3}$,

$$Q = \begin{bmatrix} 0.3 & 0.5 & 0.2 \\ 0.2 & 0.1 & 0.4 \\ 0.1 & 0.3 & 0.1 \end{bmatrix}, \quad O = \begin{bmatrix} 0.2 & 0.3 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.1 & 0.2 & 0.1 \end{bmatrix},$$

$$A_1 = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.2 & 0.4 & 0.5 \\ 0.1 & 0.3 & 0.4 \\ 0.2 & 0.4 & 0.2 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0.2 & 0.3 & 0.2 \\ 0.1 & 0.5 & 0.4 \\ 0.3 & 0.2 & 0.1 \\ 0.5 & 0.2 & 0.2 \end{bmatrix},$$

$$H^1 = \begin{bmatrix} -0.5 & 0.1 & 0 & 0.1 & 0.2 & 0.1 \\ 0.1 & -0.5 & 0.1 & 0.2 & 0.1 & 0 \\ 0 & 0.1 & -0.6 & 0 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0 & -0.6 & 0.2 & 0.1 \\ 0.2 & 0.1 & 0.3 & 0.2 & -0.9 & 0.1 \\ 0.1 & 0 & 0.2 & 0.1 & 0.1 & -0.5 \end{bmatrix},$$

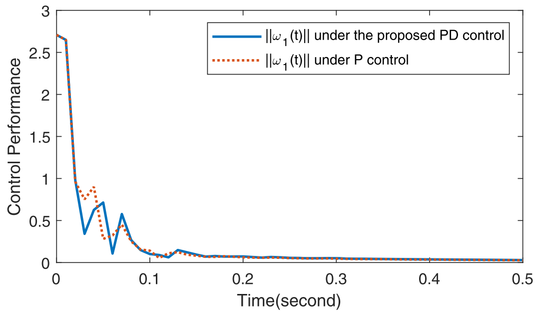


Fig. 3. Comparative simulations of PD control versus P control: Evolutions of $||\boldsymbol{\omega}_1(t)||$.

$$H^2 = \begin{bmatrix} -0.7 & 0.3 & 0 & 0.2 & 0.1 & 0.1 \\ 0.3 & -0.7 & 0.2 & 0.1 & 0.1 & 0 \\ 0 & 0.2 & -0.9 & 0 & 0.4 & 0.3 \\ 0.2 & 0.1 & 0 & -0.6 & 0.2 & 0.1 \\ 0.1 & 0.1 & 0.4 & 0.2 & -0.9 & 0.1 \\ 0.1 & 0 & 0.3 & 0.1 & 0.1 & -0.6 \end{bmatrix},$$

$$H^3 = \begin{bmatrix} -0.6 & 0.2 & 0 & 0.2 & 0.1 & 0.1 \\ 0.2 & -0.8 & 0.3 & 0.1 & 0.2 & 0 \\ 0 & 0.3 & -0.9 & 0 & 0.4 & 0.2 \\ 0.2 & 0.1 & 0 & -0.7 & 0.3 & 0.1 \\ 0.1 & 0.2 & 0.4 & 0.3 & -1.1 & 0.1 \\ 0.1 & 0 & 0.2 & 0.1 & 0.1 & -0.5 \end{bmatrix},$$

$$\Phi^1 = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.7 \end{bmatrix}, \Phi^1 = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.7 \end{bmatrix},$$

$$\Phi^3 = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.6 & 0 \\ 0 & 0 & 0.4 \end{bmatrix}.$$

Taking $\mu_r = 0.5$, it is obvious that

$$|f_r(\chi_1) - f_r(\chi_2)| \le 0.5 \cdot |\chi_1 - \chi_2|, \tag{25}$$

for any $\chi_1, \chi_2 \in \mathbb{R}$.

By setting the PD controller parameters to $b_p^1 = 2, b_p^2 = 2, b_p^3 = 20$, and using the MATLAB YALMIP Toolbox, a feasible solution for the LMI (7) can be solved as

$$Z = I_6 \otimes \begin{bmatrix} 1.2458 & 1.3782 & -0.6398 \\ -1.0886 & -1.8741 & -0.7216 \\ 0.4556 & 0.6692 & 2.0635 \\ 1.0509 & 0.5792 & 0.8720 \end{bmatrix}.$$

According to Theorem 2, the passivity of the network (24) is achieved via the PD control (5).

Similarly, letting $b_p^1 = 15, b_p^2 = 34, b_p^3 = 27$, and using the MATLAB YALMIP Toolbox, the following matrices can be found

$$Z = I_6 \otimes \begin{bmatrix} -0.0616 & -0.5053 & 0.4636 \\ -0.3059 & 0.9774 & 1.2215 \\ 0.7842 & 1.0227 & 0.8224 \\ 2.9195 & 0.0228 & 0.2124 \end{bmatrix},$$

and

$$E_1 = I_6 \otimes \begin{bmatrix} 0.2610 & -0.1718 & -0.1664 \\ -0.1718 & 0.3499 & 0.3966 \\ -0.1664 & 0.3966 & 0.5786 \end{bmatrix},$$

satisfying (14). From Theorem 2, the input-strict passivity of the network (24) is achieved via the PD control (5).

Finally, if the controller parameters are set $b_p^1 = 15, b_p^2 = 30, b_p^3 = 25$, then a set of feasible solutions for the LMI (15) are obtained as

$$Z = I_6 \otimes \begin{bmatrix} 46.7757 & -8.4423 & -3.5045 \\ -20.2481 & 4.3914 & 4.0261 \\ -13.9148 & 0.4410 & 1.5961 \\ 8.0193 & 1.2771 & -3.2751 \end{bmatrix},$$

and

$$E_2 = I_6 \otimes \begin{bmatrix} 0.0100 & 0 & 0 & 0 \\ 0 & 0.0100 & 0 & 0 \\ 0 & 0 & 0.0100 & 0 \\ 0 & 0 & 0 & 0.0100 \end{bmatrix},$$

which means that the output-strict passivity of the network (24) is achieved via the PD controller (5). On this basis, the proposed PD controller demonstrates superior oscillation suppression and convergence acceleration in FOCNNMWs, as evidenced by Fig. 2(a)'s comparative analysis with uncontrolled systems. Under external inputs, output norms rapidly stabilize (e.g., $||\boldsymbol{\beta}_i(t)|| < 2$ within 0.3s, $i = 1, 2, 3, 4, 5, 6$) while maintaining bounded energy dissipation, validating robust disturbance rejection—in contrast to the persistent oscillation ($||\boldsymbol{\beta}_2(t)|| > 2$ when $t = 0.3$s) observed in uncontrolled cases.

Select $b_D^1 = 0.4$, $b_D^2 = 0.15$, $b_D^3 = 0.25$. Then, it is easy to straightforward that $A_1^T A_1 > 0$ and (16) hold. correspondingly, under the zero-input condition (i.e., $\boldsymbol{\alpha}_i(t) = 0$), the PD-controlled network (solid line) demonstrates superior performance (see Fig. 2(b)): the output norm $||\boldsymbol{\omega}_i(t)||, i = 1, 2, 3, 4, 5, 6$ converges to near-zero (< 0) within 2s, with error norm decreasing from 3.0 to 0.5 in just 0.3s. In contrast, the uncontrolled network in Liu and Wang (2021) (dashed line) exhibits persistent oscillation (with certain $||\boldsymbol{\omega}_i|| > 0.3$ when $t > 2$s) over 8s. This rapid convergence aligns with Theorem 4's LMI condition, confirming the synchronization under zero input. To further validate the structural advantage of the derivative term, a comparative simulation between pure proportional (P) control and the proposed PD control is conducted under identical network conditions. The evolution of the synchronization error norm $||\omega_1(t)||$ in Fig. 3 clearly demonstrates the performance superiority of PD control in both convergence speed and dynamic coordination. Quantitatively, the PD-controlled system exhibits a faster initial response, with $||\omega_1(t)||$ beginning its steep decline around $t = 0.3$s, followed by a brief transient rebound near $t = 0.4$s due to the derivative-induced phase lead. In contrast, the P-controlled response decays more gradually, lacking the rapid initial decline characteristic of PD control. The PD controller subsequently drives the error into and maintains it within a narrow bound ($||\omega_1(t)|| < 0.05$) significantly earlier than the P controller. This accelerated convergence, coupled with the effective damping of multi-weight coupling oscillations, confirms that the derivative term provides essential dynamic coordination, making PD control more suitable for synchronizing fractional-order multi-weighted networks despite its slightly more pronounced transient behavior.

Remark 6. The simulation results explicitly validate the disturbance rejection capability inherent in the proposed passivity-based control framework. In Figs. 1(a) and 2(a), the system is subjected to persistent time-varying inputs $\boldsymbol{\alpha}_i(t)$, representing bounded external disturbances. The evolution of $||\boldsymbol{\beta}_i(t)||$ shows that although the input energy is continuously injected, the output remain bounded and exhibit a convergent tendency. This behavior aligns with the passivity inequality, confirming that the supplied disturbance energy is dissipated through the network and controller, preventing instability. Furthermore, when the inputs are removed (Figs. 1(b) and 2(b)), the synchronization errors converge to zero exponentially, highlighting the system's ability to recover completely once disturbances cease. These observations collectively demonstrate that the PD controllers not only achieve passivity and synchronization under ideal conditions but also ensure robust performance against persistent disturbances and model uncertainties.

Remark 7. The proposed passivity and synchronization criteria for the FOCNNMWs (2) in Theorems 1–4 are formulated as matrix inequalities whose dimensions scale with both network size (node count) and state-space dimensionality. This inherent scaling property introduces computational challenges when dealing with large-scale networks. However, through the application of the Schur complement lemma, these matrix inequalities can be systematically transformed into standard LMIs. This transformation enables efficient numerical solutions using MATLAB's LMI Toolbox, thereby overcoming the direct computational limitations of the original formulation. The methodological conversion serves as a crucial bridge between theoretical conditions and practical implementation, ensuring the proposed criteria remain computationally tractable for empirical validation despite their structural complexity.

Remark 8. Notably, LMI framework addresses the analysis of passivity and synchronization in complex multi-weight networks by systematically reformulating intricate stability conditions into a numerically tractable set of convex constraints. This contrasts with conventional algebraic or frequency-domain approaches, which often struggle to scale efficiently for high-dimensional, multi-weight dynamics. Beyond this structural strength, the LMI approach offers decisive computational utility: the conditions derived in Theorems 1–4 are directly solvable using established numerical tools such as MATLAB's YALMIP, transforming theoretical criteria into readily verifiable and implementable controller designs–a clear advantage over methods that merely yield nonconstructive or existential results.

Remark 9. The passivity analysis of integer-order CNNs with time delays was conducted in Wang et al. (2017) and Zhang et al. (2018) through the application of Lyapunov-Krasovskii functionals. However, this approach exhibits inherent limitations when extended to fractional-order systems primarily due to the non-commutative properties of fractional calculus operators. As shown in Wang et al. (2017), the Lyapunov-Krasovskii function is constructed as

$$V(t) = \sum_{i=1}^{n} z_i^T(t) M z_i(t) + \sum_{s=1}^{m} c_s \int_{t-\tau_s}^{t} z_i^T(h) P_s z_i(h) dh,$$

in which $z_i(t)$ denotes the synchronization error in the considered CNNs, $0 < c_s$ (gain) and $0 < \tau_s$ (delay) are design parameters in the $s-$th coupling topology, M and P_s are positive definite matrices. The $\eth-$th order derivative of the term $\sum_{s=1}^{m} c_s \int_{t-\tau_s}^{t} z_i^T(h) P_s z_i(h) dh$ constitutes an infinite series that presents significant computational intractability. This complexity escalates substantially when considering time-varying delays τ_s, where the derivative operation on the integral term $\sum_{s=1}^{m} c_s \int_{t-\tau_s}^{t} z_i^T(h) P_s z_i(h) dh$ becomes more intricate due to the coupled effects of fractional differentiation and delay variability. These analytical challenges necessitate the development of specialized Lyapunov-Krasovskii functionals that can systematically address the compounded complexities arising from the interplay between fractional-order dynamics and time-varying delay effects, while enabling feasible mathematical conditions for passivity and synchronization analysis.

5. Conclusions

This paper investigates the passivity and synchronization of FOCNNMWs under the PD control strategy, establishing theoretical criteria through Lyapunov methods, LMIs, and Mittag-Leffler function analysis. It is validated numerically that proposed PD controllers ensure passivity (including input-strictly and output-strictly variants) and synchronization under zero-input conditions. Unlike prior integer-order studies, the framework is limited to fractional-order systems, which can treat the former as its special case. While the conclusions are specifically derived for PD control, the developed analytical framework is versatile and provides a foundation for broader applications. This opens several directions for future research, such as extending the methodology to other control strategies (e.g., PI or event-triggered control), studying networks under more complex dynamics, and applying the results to practical domains like secure communication. Additionally, theoretical improvements to the LMI-based criteria themselves, aimed at reducing conservatism, constitute another important avenue.

CRediT authorship contribution statement

Yu Li: Writing – original draft, Methodology, Formal analysis; **Xiulan Zhang:** Writing – review & editing, Validation, Supervision; **Jiancheng Zhang:** Writing – review & editing, Writing – original draft, Validation, Conceptualization; **Jianwei Er:** Writing – review & editing, Methodology; **Jinde Cao:** Writing – review & editing, Writing – original draft, Conceptualization; **Heng Liu:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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